Reactome Data Model Glossary

Database-Derived Edition | V95

All class descriptions, attribute definitions, and relationship descriptions derived from live Neo4j graph database queries, informed by the V90 Reactome Data Model Glossary.

Overview

Reactome is a slot-and-frame knowledge representation of a network of molecular transformations mediated by gene products. Classes (frames) are categories like Reaction and Complex. Related classes are grouped into superclasses to allow inheritance of shared attributes. For example, the PhysicalEntity superclass has child classes such as Complex and GenomeEncodedEntity. Classes have slots (attributes) whose values define properties of instances.

"Defining" attributes are used in the data model to distinguish instances of a class, ensuring that identical information — for example, a water molecule localised to the cytosol, or a reaction by which a particular molecule is translocated from the extracellular space into the cytosol by a particular transporter — is represented exactly once in Reactome. Many attributes of a class instance are themselves instances of other classes.

In this glossary, attribute names appear in italics. Class names appear in bold. Superclass labels marked [superclass] are abstract grouping labels not used for manual annotation. Mixin labels marked [mixin] are structural graph co-labels not visible in the curator tool.

Section 1 — Database Infrastructure Classes

These classes maintain database integrity and support audit, versioning, and deletion workflows. None are used for biological annotation.

DatabaseObject *[superclass]*

The root superclass of every instance in the Reactome database. Provides the core identity and display properties inherited by every other class. Not used for manual annotation.

Attribute / Relationship	Description
dbId	A unique internal database identifier automatically assigned to every new instance when it is first committed to the central database. Must never be modified or manually created by a curator.
schemaClass	The name of the concrete class of this instance (e.g., 'Reaction', 'Complex'). Set automatically by the curator tool; not edited manually.

displayName	The human-readable name displayed on the Reactome website. Auto-generated by the curator tool from other attribute values. Curators do not set this directly.
stId	The Stable Identifier for this instance. Format: R-{species code}-{dbId} where species codes include HSA (Homo sapiens), MMU (Mus musculus), ALL (species-independent), NUL (multi-species or unspecified). Once assigned at first release, stId never changes.
stIdVersion	The Stable Identifier including a version suffix (e.g., R-HSA-123456.3). The version number Z is incremented each time a substantive change is made between database releases.
name	A short textual description of the instance. For Events and PhysicalEntities this is a defining attribute that must be unique within the class.
isInDisease	Boolean. True for instances that are part of disease pathway annotations.
isInferred	Boolean. True for instances computationally inferred from curated instances in another species during the cross-species inference pipeline.
releaseDate	The date this instance first appeared in a public Reactome release.
lastUpdatedDate	The date this instance was last substantively modified.

Trackable *[mixin]*

A structural mixin co-label applied to all curated instances (PhysicalEntities, Events, Regulations) that participate in Reactome version-tracking. Not visible in the curator tool; not used for manual annotation. Enables the StableIdentifier versioning system and the UpdateTracker change-history machinery.

Deletable *[mixin]*

A structural mixin co-label applied alongside Trackable to all curated instances that can be deleted through the standard deletion workflow. Not visible in the curator tool; not used for manual annotation. Marks an instance as eligible for the Deleted audit-record workflow: when a curator deletes such an instance, a Deleted record is automatically created preserving the deletion rationale and replacement links.

MetaDatabaseObject *[superclass — mixin]*

A co-label grouping all database-management objects that describe the database itself rather than its biological content. Applied to four concrete classes: UpdateTracker, DeletedInstance, Deleted, and Release. Allows database-management queries to be scoped away from biological content. Not used for manual annotation.

InstanceEdit

Records the date, time, and responsible person whenever a Reactome instance is created or modified. Generated automatically when the central database is updated from the curator tool. Must not be manually created or edited.

Attribute / Relationship	Description

dateTime	A timestamp automatically generated when the central database is updated from a curator tool project. Do not edit manually.
author → Person	The curator or developer responsible for the database change recorded by this InstanceEdit. Also used for authors of LiteratureReferences.
created (outgoing)	Records which instance was created or first committed in this edit. Spans all biological classes.
authored (outgoing to Event or PhysicalEntity)	Records the intellectual authorship of the biological content of an event or entity. Distinct from 'created': authored captures who wrote the biological annotation; created captures the database commit. Populates the displayed author information on Reactome pathway pages.
edited (outgoing to Event)	Records subsequent editorial modification of an event's biological content.
reviewed (outgoing to Event)	Records that an event underwent external expert review. Populates the displayed reviewer information on Reactome pathway pages.
revised (outgoing to Event)	Records that an event was revised following external expert review.
internalReviewed (outgoing to Event)	Records that an event underwent internal Reactome review.
structureModified (outgoing to Pathway)	Records that the structural composition of a pathway — its hasEvent membership — was changed.

Deleted

Created automatically when a curator deletes a curated instance. Preserves a permanent audit trail of what was removed, why it was removed, and what replaced it. Previously documented as '_Deleted'.

Attribute / Relationship	Description
deletedInstanceDbId	The internal database identifier (dbId) of the removed instance, preserved for traceability.
curatorComment	An optional free-text comment explaining the reason for the deletion.
reason → DeletedControlledVocabulary	A controlled vocabulary term describing why the instance was deleted.
deletedInstance → DeletedInstance	Points to the companion DeletedInstance record preserving the class name, stable identifier, display name, and other identifying metadata of the removed instance.
replacementInstances (outgoing)	Points to replacement instance(s) in the live database when the deleted instance was replaced rather than simply removed — used for merges, splits, and superseded annotations.

DeletedControlledVocabulary

A ControlledVocabulary subclass providing the controlled list of reasons for which an instance may be deleted. Populates the reason attribute of a Deleted record. Maintained by the Editor-in-Chief. Current terms:

Attribute / Relationship	Description
Duplicate	The instance was a duplicate of an existing instance and has been removed in favour of the original.

Killed not replaced	The instance was found to be invalid and has been removed without any replacement.
Merged	The instance has been merged with one or more other instances into a single consolidated record.
Obsoleted	The instance is no longer in use. The replacementInstances attribute or curatorComment indicates the current equivalent.
Split	The instance has been split into two or more separate instances.
Literature_retracted	The instance was based on literature that has since been formally retracted.
never_released_not_needed	The instance was never publicly released and does not require a formal deletion record.

DeletedInstance

A companion record to Deleted preserving key identifying metadata about a deleted instance so it can be traced after removal. Always linked from a parent Deleted record. Not created or edited directly by curators. Previously documented as '_DeletedInstance'.

Attribute / Relationship	Description
clazz	The schema class name of the deleted instance (e.g., 'EntityWithAccessionedSequence').
deletedInstanceDbId	The original internal database identifier of the deleted instance.
deletedStId	The stable identifier at time of deletion, if one had been assigned.
name	The display name of the deleted instance at time of deletion.

Release

Records a single Reactome database release: its sequential release number and publication date. Maintained by the Editor-in-Chief and Managing Editor; curators do not create or modify Release instances. Previously documented as '_Release'.

Attribute / Relationship	Description
releaseNumber	A sequential integer identifying the release (e.g., 94, 95).
releaseDate	The date this release was made publicly available.
release ← UpdateTracker	Each UpdateTracker recording a change published in this release points back to this Release node.

UpdateTracker

Records the specific changes made to a curated instance between one Reactome release and the previous one. Generated automatically by the UpdateTracker script at release time. Must not be manually edited. Previously documented as '_UpdateTracker'.

Attribute / Relationship	Description
action	A list of strings describing the type(s) of change. Common values: addName (a new name was added), updateContainedComponent (complex component list changed), modifyName (existing name altered), modifyText (summation text

	updated), updateContainedRLE (RLE within a pathway added or removed), modifyEndCoordinate / modifyStartCoordinate (EWAS coordinate changed), add_removeHasComponent (complex component added or removed), add_removeReferenceEntity (reference entity changed), addLiteratureReference (new literature reference added).
updatedInstance (outgoing)	Points to the curated instance — Event, PhysicalEntity, Regulation, etc. — that was changed.
release → Release	Points to the Release in which this change was published.

DBInfo

A singleton class — exactly one instance exists in the database at any time — recording metadata about the current graph state. Also carries the Release label. Generated and maintained automatically at each release; curators do not edit it.

Attribute / Relationship	Description
name	Always 'reactome'. Identifies this as the Reactome database.
releaseNumber	The current release number.
releaseDate	The date of the current release.
checksum	A database integrity checksum used by the release pipeline.
neo4j	The version of Neo4j used for this release build.

Section 2 — Event Classes

Events describe biological transformations. The Event superclass has two primary subclass families: Pathway (multi-step) and ReactionLikeEvent (single-step). The compartment of an event involving participants in more than one compartment should reflect all compartments of its participants.

2.1 Shared Event Attributes and Relationships

All Event subclasses share the following in addition to DatabaseObject attributes.

Attribute / Relationship	Description
species → Species	The biological species in which this event is annotated as occurring. Almost always Homo sapiens for human Reactome curation. For chimeric cross-species inference reactions, all involved species are listed. In host-pathogen interactions, the host species is listed here (see relatedSpecies for the pathogen).
summation → Summation	A free-text description of the event. Citations in the summation provide background context only — primary experimental evidence must be linked directly to the event via literatureReference, not via the summation.
literatureReference → LiteratureReference or Book	One or more publications providing direct experimental evidence for this event. For reactions, these must demonstrate the specific reaction in humans (or in a closely related species for chimeric reactions). Multiple references are entered individually.
compartment → Compartment	The subcellular location(s) of the event. Use the Compartment vocabulary: 'nucleoplasm' not 'nucleus'; 'cytosol' not 'cytoplasm'. For events spanning multiple compartments, list all relevant terms.
goBiologicalProcess → GO_BiologicalProcess	A Gene Ontology Biological Process term describing the biological process this event represents. Primarily assigned to Pathway instances.

evidenceType → EvidenceType	A GO evidence code indicating the type of evidence. Set automatically to IEA on computationally inferred events. See the EvidenceType entry for the one code curators may set manually.
inferredTo (outgoing)	Points to the orthologous event(s) in other species computationally inferred from this curated event.
precedingEvent → ReactionLikeEvent or Pathway	The event that immediately precedes this one — typically the event whose output is consumed as input by the present event. When linking across pathway boundaries, always link to a specific terminal ReactionLikeEvent, not to a Pathway node.
regulatedBy → any Regulation subclass	One or more Regulation instances describing positive or negative regulation of this event.
crossReference → DatabaseIdentifier	An optional reference to this event in an external database.
figure → Figure	An optional illustrative figure associated with this event.
reviewStatus → ReviewStatus	The review status of this event — the level of internal and external review it has undergone.
previousReviewStatus → ReviewStatus	The review status before the most recent change.
disease → Disease	The disease associated with this event, for disease pathway annotations. Required on all FailedReaction instances.
relatedSpecies → Species	The 'bystander' species in host-pathogen or symbiotic interactions. The host species is in the species attribute; the pathogen/symbiont is in relatedSpecies. Example: for 'flagellin of E. coli binds human TLR5', species = Homo sapiens and relatedSpecies = Escherichia coli.
catalystActivityReference → CatalystActivityReference	Links a reaction directly to publication(s) evidencing its catalytic activity — finer-grained citation than the top-level literatureReference.
regulationReference → RegulationReference	Links a reaction directly to publication(s) evidencing a specific regulatory relationship.
requiredInputComponent → any PhysicalEntity	A PhysicalEntity that must be present for this event to occur but is not a stoichiometric input. Use only for non-trivial requirements not already expressed in the input list.
negativePrecedingEvent → NegativePrecedingEvent	Flags a computationally inferred precedingEvent connection to another RLE that a curator has reviewed and rejected.
entityOnOtherCell → any PhysicalEntity	A PhysicalEntity on a neighbouring cell that participates in a cell-cell interaction. Used in cell-cell communication pathway annotations.
reactionType → ReactionType / DrugActionType	Classifies the mode of drug action in reactions involving a Drug subclass PhysicalEntity. Terms from IUPHAR/Guide to Pharmacology.
reverseReaction → Reaction	Points to the thermodynamically reversible counterpart of this reaction, when curated.
normalPathway → Pathway	For disease pathway variants only: points to the corresponding wild-type pathway from which this disease variant was derived.
authored ← InstanceEdit	Incoming. Records who was responsible for the intellectual authorship of this event's biological content.
edited ← InstanceEdit	Incoming. Records subsequent editorial modifications of this event.
reviewed ← InstanceEdit	Incoming. Records external expert review.

revised ← InstanceEdit	Incoming. Records revision following external review.
internalReviewed ← InstanceEdit	Incoming. Records internal Reactome review.
structureModified ← InstanceEdit	Incoming. Records changes to the pathway's hasEvent composition.

Event *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Any transformation of input entities to output entities in one step (ReactionLikeEvent) or two or more steps (Pathway).

Pathway

A sequence of two or more causally connected ReactionLikeEvents and/or sub-Pathways, identified as hasEvent attributes of the pathway being annotated. A concrete example is the Reactome Apoptosis pathway, which consists of subpathways 'Caspase activation via extrinsic apoptotic signalling pathway', 'Intrinsic pathway for Apoptosis', 'Apoptotic execution phase', and 'Regulation of Apoptosis'.

Attribute / Relationship	Description
hasDiagram	Boolean. True if this pathway has a manually drawn diagram in the Reactome Pathway Browser.
hasEHLD	Boolean. True if this pathway has an Enhanced High-Level Diagram.
diagramHeight / diagramWidth	Pixel dimensions of this pathway's diagram, auto-populated from the diagram file.
maxDepth	The maximum depth of this pathway's event hierarchy, auto-calculated.
hasEvent → ReactionLikeEvent or Pathway	The constituent events of this pathway in biological order. Can contain any mixture of Reaction, BlackBoxEvent, FailedReaction, Polymerisation, Depolymerisation, CellDevelopmentStep, and sub-Pathway instances.
hasEncapsulatedEvent → Pathway	A pathway displayed within this pathway's diagram for context, but not a structural child in the event hierarchy.
goBiologicalProcess → GO_BiologicalProcess	The GO Biological Process term representing this pathway's biological process. Verified via OLS before assignment.
normalPathway → Pathway	For disease pathway variants only: points to the corresponding wild-type pathway.

TopLevelPathway

A specific kind of Pathway covering a broad topic of cellular biology. TopLevelPathways represent the highest level of the hierarchical organisation of events in Reactome, as displayed in the Pathway Browser. The single FrontPage instance maintained by the Editor-in-Chief and Managing Editor lists all TopLevelPathways. Managed exclusively by the Editor-in-Chief; not created by individual curators.

CellLineagePath

A CellLineagePath, organisationally similar to the Pathway subclass, is composed of other CellLineagePath instances or CellDevelopmentStep instances as subevents connected through shared input/output Cell instances. Both CellLineagePath and CellDevelopmentStep are associated with a GO biological process term and with a tissue term from UBERON.

Attribute / Relationship	Description
hasEvent → CellDevelopmentStep or CellLineagePath	The constituent developmental steps or nested lineage sub-paths.
hasEncapsulatedEvent → CellLineagePath	A sub-path displayed within this path's diagram for context, not a structural child.

ReactionLikeEvent *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. A molecular process in which one or more input PhysicalEntities are transformed in a single step into output PhysicalEntities, optionally mediated by a catalystActivity and subject to Regulation. Transformations include chemical reactions, translocation of molecules between compartments, association of molecules to form complexes and polymers, dissociation of complexes and polymers, and developmental/differentiation relationships among Cell instances.

Attribute / Relationship	Description
isChimeric	Boolean. True when this reaction involves participants from more than one species, used to flag reactions constructed from multi-species evidence for cross-species inference. Non-human protein names in chimeric reactions use initial-capitalisation only (e.g., Jak2); human proteins use full capitals (e.g., JAK2).
category	An automatically assigned string categorising the reaction mechanism: transition (enzymatic chemical transformation), binding (complex formation), dissociation (complex disassembly), omitted (steps not explicitly curated), or uncertain.
input → any PhysicalEntity	Each PhysicalEntity consumed in this reaction, entered individually. Include all stoichiometric participants: cofactors, cosubstrates, and small molecules such as ATP alongside the primary substrates.
output → any PhysicalEntity	Each PhysicalEntity produced in this reaction, entered individually. Include all byproducts: ADP, Pi, and other small molecule products alongside the primary products.
catalystActivity → CatalystActivity	The CatalystActivity instance describing the catalytic entity and its molecular function. A reaction may have at most one CatalystActivity per molecular function; if the catalyst enables multiple distinct functions in this reaction, a separate CatalystActivity is created for each.
regulatedBy → any Regulation subclass	One or more Regulation instances. See the Regulation class family (Section 5).
entityFunctionalStatus → EntityFunctionalStatus	Required on all FailedReaction instances. Describes the variant entity and molecular basis of the functional change.
normalReaction → ReactionLikeEvent	Required on FailedReaction instances. Points to the wild-type counterpart reaction.

Reaction

The transformation of input physical entities into output ones in a single step. Transformations include chemical changes (as in metabolism), transport of an entity between locations, association of entities to

form a complex, and dissociation of complexes. These transformations can be catalysed and regulated. Reaction is the most common ReactionLikeEvent subclass.

BlackBoxEvent

Converts inputs to outputs of a multi-step biological process in which the intervening steps are not annotated, either because the molecular details are incompletely known or because the steps do not need to be specified for the purposes of the module being curated. Examples: transcription and translation of a gene into a protein; protein degradation; transport between organelles. A fully annotated sequence of events for a BBE may be depicted in full elsewhere in Reactome.

> *A BlackBoxEvent requires explicitly specified inputs AND outputs. The 'black box' refers only to the unresolved mechanism — not to unknown participants. Do not annotate events where participants cannot be specified.*

FailedReaction

A FailedReaction instance is a step in a disease process directly affected by a loss-of-function (LoF) mutation, germline or somatic. It has its normal ReactionLikeEvent counterpart (the reaction mediated by the un-mutated gene product) specified at its normalReaction attribute, and is represented as having inputs (the abnormal PhysicalEntity plus all wild-type entities that are inputs in the normal reaction) but no outputs. FailedReaction instances are associated with disease term(s) and with an EntityFunctionalStatus record.

> *Best practice: clone the wild-type reaction, switch its type to FailedReaction, replace the wild-type entity with the variant entity, then add the disease, entityFunctionalStatus, and normalReaction attributes.*

Attribute / Relationship	Description
normalReaction → Reaction or BlackBoxEvent	The wild-type ReactionLikeEvent counterpart. Required.
entityFunctionalStatus → EntityFunctionalStatus	Describes the variant entity and the molecular basis of the functional change. Required on all FailedReaction instances.
disease → Disease	The Disease Ontology term identifying the associated disease. Required.
input → any PhysicalEntity	The variant entity plus all wild-type co-inputs from the normal reaction. No output relationship is present on FailedReaction instances.
catalystActivity → CatalystActivity	Present when the catalyst itself is the variant component that fails.
crossReference → DatabaseIdentifier	Used to cross-reference to external disease variant databases.

Polymerisation

A subclass of ReactionLikeEvent in which two or more identical molecules or complexes are assembled into a Polymer. Follows the pattern: repeating unit(s) → Polymer. The output is typically a Polymer PhysicalEntity instance.

Depolymerisation

A subclass of ReactionLikeEvent used to annotate depolymerisation, following the reverse pattern of Polymerisation: Polymer → Polymer + Unit. The input is typically a Polymer PhysicalEntity and the outputs include released units and associated small molecules.

CellDevelopmentStep

A CellDevelopmentStep is a ReactionLikeEvent containing Cell instances as its inputs and outputs. The input represents the cell of origin (starting point of the developmental event) and the output represents the destination cell type (outcome of the event). Additional attributes include regulators (molecules promoting or inhibiting the step) and requiredInputComponent (input cell biomarkers required for the action of regulators). Also associated with a relevant GO biological process term and a tissue term from UBERON.

Attribute / Relationship	Description
input → Cell	The originating cell type or state. Exactly one Cell input per CellDevelopmentStep.
output → Cell	The resulting cell type or state. Exactly one Cell output per CellDevelopmentStep.
regulatedBy → PositiveRegulation or NegativeRegulation	Molecules promoting or inhibiting this developmental transition.
requiredInputComponent → EWAS or CandidateSet	Input cell biomarkers that must be present for the regulatory molecules to act.
precedingEvent → CellDevelopmentStep	The preceding developmental step in the lineage trajectory.

2.2 Event Support Classes

NegativePrecedingEvent

Used to flag an Event that should not be annotated as a value of another Event's precedingEvent attribute. Records a computationally inferred precedingEvent connection between two RLEs that a curator has reviewed and rejected. The NegativePrecedingEvent record prevents the erroneous link from being re-inferred at subsequent releases.

Attribute / Relationship	Description
precedingEvent → Reaction or BlackBoxEvent	The upstream RLE whose inferred connection to the current event is being rejected.
reason → NegativePrecedingEventReason	A controlled vocabulary term explaining why the connection was rejected.

NegativePrecedingEventReason

A subclass of ControlledVocabulary used to indicate why a computationally inferred causal connection (precedingEvent) between two ReactionLikeEvents was manually rejected by a curator. Current terms:

Attribute / Relationship	Description
Abundant Molecules	The two reactions share a participant present abundantly in the cell (e.g., ATP, water), creating a spurious inferred connection between reactions that are not causally linked.

Different Cell Type	The two reactions occur in different cell types; their outputs cannot physically connect.
Drug molecule	The inferred connection runs through a drug molecule. Only physiologically occurring entities should generate inferred precedingEvent links.
Futile Cycle	The two reactions form a thermodynamically futile cycle that counteracts each other, so no meaningful causal direction exists between them.
Incorrect Order	The inferred ordering of the two reactions is biologically implausible given current pathway knowledge.
Negative relationship	The upstream reaction inhibits rather than enables the downstream one, making a precedingEvent annotation mechanistically misleading.

Summation

A free text description of an Event or PhysicalEntity. Citations that provide useful background information but are not sources of primary data for the event can be linked to summations via the literatureReference attribute. LiteratureReferences providing primary experimental evidence for the details of an event should be linked directly to the event, not to its summation.

Attribute / Relationship	Description
text	The free-text description of the event or entity. May include HTML markup for web display.
literatureReference → LiteratureReference	Papers supporting the text description but NOT the primary evidence for the event. These appear as hyperlinked citations on the Reactome website.

ReviewStatus

A ControlledVocabulary instance recording the level of internal and external review an Event has undergone. Controls whether the event is included in public releases (three stars and above). Set by the Managing Editor during the release process; curators must not change the review status of released events without Managing Editor approval.

Attribute / Relationship	Description
one star	Structure updated after internal review. NOT RELEASED publicly.
two stars	Structure updated after external review. NOT RELEASED publicly.
three stars	Internally reviewed; awaiting external review. RELEASED.
four stars	Restructured after external and internal review. RELEASED.
five stars	Externally reviewed and approved without requiring structural changes. RELEASED.

Figure

Figures for Reactome events are created in collaboration with Reactome's graphic illustrator. Curators needing to add a figure should consult with the illustrator and enter plans into the illustration calendar.

Attribute / Relationship	Description
url	The path to the image file on the server. Format: /figures/.../{filename}.svg or .jpg.

Section 3 — Physical Entity Classes

A PhysicalEntity is a physical substance that can interact with other substances. PhysicalEntities include all kinds of small molecules, proteins, nucleic acids, chemical compounds, complexes, larger macromolecular assemblies, atoms (including ionised atoms), electrons, and photons. PhysicalEntities can be inputs, outputs, catalysts, regulators, or requirements in ReactionLikeEvents.

3.1 Shared PhysicalEntity Attributes and Relationships

Attribute / Relationship	Description
compartment → Compartment	The subcellular location of this PhysicalEntity. Required on most subclasses. If a PhysicalEntity occurs in more than one compartment, different instances are created for each location, distinguished by their compartment attribute. Use the Compartment vocabulary: 'nucleoplasm' not 'nucleus'; 'cytosol' not 'cytoplasm'.
species → Species	The biological species of this entity. Not assigned to species-independent molecules (e.g., ATP occurring in many species is not assigned a species).
summation → Summation	An optional free-text description of this entity.
literatureReference → LiteratureReference	Optional literature references evidencing the existence or properties of this specific entity instance.
crossReference → DatabaseIdentifier	An optional cross-reference to this entity in an external database (e.g., KEGG, IntAct), in addition to the primary referenceEntity link.
disease → Disease	Associates this entity with a disease. Applied to variant PhysicalEntities — mutant proteins, disease-associated complexes — in disease pathway annotations.
cellType → CellType	An optional Cell Ontology term restricting this entity to a specific cell type when its modification state, localisation, or behaviour is cell-type-specific. Applied to EWAS, SimpleEntity, Complex, ChemicalDrug, and OtherEntity instances.
goCellularComponent → GO_CellularComponent	A supplementary GO Cellular Component term used when the standard Compartment vocabulary lacks sufficient specificity. Used primarily in infectious disease and host-pathogen annotations for 'host cell' locations (e.g., 'host cell cytosol', 'host cell plasma membrane') not in the Compartment vocabulary.
inferredTo (outgoing)	Points to the orthologous entity in another species inferred from this entity.
isChimeric	Boolean. True for entities built from participants of more than one species.
isInDisease	Boolean. True for variant entities in disease pathway annotations.

PhysicalEntity *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. PhysicalEntity contains the classes Cell, Complex, Drug, EntitySet, GenomeEncodedEntity, OtherEntity, Polymer, and SimpleEntity.

EntityWithAccessionedSequence

The subclass of GenomeEncodedEntities that can be associated through the referenceEntity attribute with reference molecules in UniProt (proteins) or ENSEMBL (DNA, RNA) databases. An EntityWithAccessionedSequence (EWAS) represents a specific protein, RNA, or DNA molecule in a specific subcellular compartment, optionally carrying post-translational or other modifications. A protein

that has been partially purified and whose enzymatic properties are known but whose amino acid sequence has not been deposited in a reference database is a GenomeEncodedEntity but not an EWAS.

> *Naming convention: PTMs as PREFIX using single-letter codes and residue coordinates (e.g., p-S473-AKT1). Cleavage products or fragment differences as SUFFIX. Different compartments always require different EWAS instances for the same protein.*

Attribute / Relationship	Description
referenceEntity → ReferenceGeneProduct, ReferenceIsoform, ReferenceDNASequence, or ReferenceRNASequence	The canonical sequence record from UniProt or ENSEMBL to which this EWAS is linked. Required. Use ReferenceGeneProduct for the canonical UniProt protein sequence; use ReferenceIsoform only when experimental literature specifically supports the role of a non-canonical isoform.
startCoordinate	The amino acid (or nucleotide) position at which this EWAS begins. Use 1 for full-length proteins; use the UniProt coordinate for defined fragments. Default: 1.
endCoordinate	The amino acid (or nucleotide) position at which this EWAS ends. Use -1 when the full C-terminus (or 3' end) is included.
referenceType	Automatically populated string indicating the type of reference entity: ReferenceGeneProduct (for proteins), ReferenceDNASequence (for gene loci), ReferenceRNASequence (for transcripts), or ReferenceIsoform.
hasModifiedResidue → any AbstractModifiedResidue subclass	One modification record per post-translational, genetic, or nucleotide modification carried by this EWAS. See Section 4 for the modification class hierarchy.
compartment → Compartment	The subcellular location of this EWAS. Required. A separate EWAS instance must be created for each distinct compartment in which the same protein participates.

GenomeEncodedEntity

Any informational macromolecule (DNA, RNA, protein) or entity derived from one by post-synthetic modifications, for example covalently modified forms of proteins. GenomeEncodedEntity is a subclass of PhysicalEntity with one concrete subclass: EntityWithAccessionedSequence. Unlike an EWAS, a GenomeEncodedEntity is not required to have a reference entity, though a reference to an entity in a database can be specified in the crossReference attribute. Used for partially characterised gene products whose enzymatic properties are known but whose sequence is not deposited in a reference database.

Complex

A PhysicalEntity representing a larger functional unit formed by the covalent or non-covalent association of two or more PhysicalEntity components (which can themselves be Complexes or Sets). Usually complex components are proteins, but they can also be RNA or DNA molecules, or SimpleEntity chemicals such as metal ions, nucleotides, or lipids that function as enzyme cofactors. Complexes include homomultimers (e.g., ESR1 homodimer [nucleoplasm]), heteromultimers (e.g., FANCD2:FANCI [nucleoplasm]), and complexes composed of other complexes (e.g., PPARG:RXRA:Corepressor Complex [nucleoplasm]).

> *Naming: colon-separated component names in order. For growing complexes, the new component is appended to the RIGHT. Do not use the words 'complex', 'homo', or 'hetero' in names unless part of a widely established common name (e.g., RNA pol II complex, Mediator complex).*

Attribute / Relationship	Description

hasComponent → any PhysicalEntity	Each component listed individually. Multiple copies use a stoichiometry prefix (2x, 3x). The same component in different compartments must be listed as separate EWAS instances.
compartment → Compartment	The primary subcellular location of this complex. For transmembrane complexes spanning multiple compartments, the primary location (typically the membrane) is specified here. Locations of components in other compartments are auto-populated into includedLocation during the release process.
includedLocation → Compartment	Auto-populated during the database release process to record secondary compartments occupied by components of a transmembrane or multi-compartment complex. Curators should not set this attribute manually.
entityOnOtherCell → any PhysicalEntity	A component of this complex located on a neighbouring cell rather than on the same cell — used in cell-cell communication contexts.

EntitySet *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Two or more PhysicalEntities grouped because of a shared molecular feature or function. The superclass for CandidateSet and DefinedSet. While sets are by default homogeneous (members having the same PhysicalEntity subclass), they are not required to be — for example, the DefinedSet 'Platelet alpha granule contents' contains EWAS instances, Complexes, and other Sets as members.

DefinedSet

A collection of well-characterised physical entities that are functionally indistinguishable for the purpose of Reactome annotation — for example, a collection of protein isoforms that all catalyse the identical reaction, or alternative subunits that can substitute for one another in the same complex position. A DefinedSet is formally a list of instances linked by logical ORs: any member can participate in the annotated reaction in place of any other.

Attribute / Relationship	Description
hasMember → any PhysicalEntity	The confirmed, functionally equivalent members. Each entered individually. All members must share the same compartment.
compartment → Compartment	The shared subcellular location of all members. All members must be in the same compartment.
isOrdered	Boolean. If true, the position of each member corresponds to the position of a member in a paired ordered set. For example, in an ordered substrate set paired with an ordered product set, substrate1 yields only product1 and substrate2 yields only product2. Sets are considered ordered by default.

CandidateSet

A collection of physical entities that are functionally indistinguishable for the purpose of Reactome annotation, some of which are well-characterised (the 'members' of the set) and some of which are incompletely characterised (the 'candidates' of the set), as judged by the curator and the outside expert reviewers. Members are related by a logical OR: either one OR another can participate in the reaction. A CandidateSet with no confirmed members (candidates only) is valid — it represents a case where the participating molecule is inferred but not yet experimentally identified. Example: RIP2 ubiquitin ligases — one member (ITCH) has been shown to participate; the candidates are thought to do so.

> *Naming convention: confirmed members listed first, candidates in parentheses — e.g., CCR2,(CCR4,CCR6). All members and candidates must share the same compartment.*

Attribute / Relationship	Description
hasMember → any PhysicalEntity	Confirmed, well-characterised members — entities directly demonstrated to participate in the annotated reaction.
hasCandidate → any PhysicalEntity	Unconfirmed or incompletely characterised candidates — entities thought to participate based on functional similarity or inference, lacking direct experimental confirmation.
compartment → Compartment	The shared subcellular location of all members and candidates. All must be in the same compartment.
isOrdered	Boolean. Whether positional correspondence between members of paired ordered sets is meaningful.

SimpleEntity

A defined chemical species not encoded directly or indirectly in the genome — typically a small molecule such as ATP, ethanol, water, or a metal ion. The detailed structure is specified by linking to the ChEBI external database via the referenceEntity attribute. Separate SimpleEntity instances are needed for each subcellular location in which the molecule is found — for example, ATP [cytosol] and ATP [nucleoplasm] are separate instances. SimpleEntities such as ATP that occur in many species are not assigned a species attribute.

Attribute / Relationship	Description
referenceEntity → ReferenceMolecule	The ChEBI record for this molecule providing its chemical identity, formula, and name. Required.
compartment → Compartment	The subcellular location. Required. A separate SimpleEntity must be created for each distinct compartment.
crossReference → DatabaseIdentifier	Optional cross-references to this molecule in additional external databases such as KEGG COMPOUND or PubChem.

Polymer

Molecules consisting of indeterminate numbers of repeated units, and complexes containing repeated units whose stoichiometry is variable or unknown. The repeated unit(s) (identified in the repeatedUnit attribute) can be any PhysicalEntity. More than one repeatedUnit value implies the relative numbers of units are unknown. If units are in known proportions, a Complex of the appropriate number of units is used as the repeatedUnit. The size range can optionally be specified with minUnitCount and maxUnitCount. Examples: 'glycogen' with 'glucose' as repeatedUnit; 'fibrin multimer' with 'fibrin monomer' (itself a Complex) as repeatedUnit; a microtubule with a Complex of alpha and beta tubulin as repeatedUnit.

Attribute / Relationship	Description
repeatedUnit → any PhysicalEntity	The building block(s). A single value for a homo-polymer or fixed-stoichiometry polymer; multiple values when the ratio of units is unknown or variable.
minUnitCount	The minimum number of repeated units, when known. Optional.
maxUnitCount	The maximum number of repeated units, when known. Optional.

OtherEntity

PhysicalEntities that we are unable or unwilling to describe in chemical detail and which therefore cannot be placed in any other class. OtherEntity represents complex cellular structures that take part in a reaction but which cannot or should not be defined at the molecular level. Example 1: Cell membrane — where protein X associates with the membrane but the actual component with which it interacts is unknown. Example 2: Kinesin-1 binds to mitochondria, secretory vesicles, and parts of the Golgi — these complicated structures are OtherEntities. Example 3: Holliday structure [nucleoplasm].

Attribute / Relationship	Description
compartment → Compartment	The subcellular location of this structure. Required.
crossReference → DatabaseIdentifier	Optional cross-reference to an external database record or Reactome stable identifier describing this structure.

Cell

A subclass of PhysicalEntity representing a type of cell in a particular state of development or differentiation. Cell instances have cellType, tissueLayer, tissue, and organ as attributes populated using terms from the Cell Ontology and UBERON. Cell instances also have multi-valued proteinMarker and RNAMarker attributes identifying molecules specific to the cell type or state, each associated with literature references via MarkerReference instances.

Attribute / Relationship	Description
cellType → CellType	A Cell Ontology term identifying the type of cell. Required.
tissue → Anatomy	A UBERON term identifying the tissue in which this cell resides.
organ → Anatomy	A UBERON term identifying the organ in which this cell resides.
tissueLayer → Anatomy	A UBERON term identifying the tissue layer in which this cell resides.
species → Species	The biological species of this cell type.
proteinMarker → EWAS	A protein identified as a molecular marker specific to this cell type or state. Supported by a MarkerReference instance.
RNAMarker → EWAS	An RNA molecule identified as a molecular marker specific to this cell type or state. Supported by a MarkerReference instance.
markerReference → MarkerReference	One MarkerReference instance per marker, linking the marker EWAS to the publication(s) supporting its use as a marker for this Cell.

3.2 Drug Classes

The Drug superclass and its three subclasses represent therapeutic substances administered to alter metabolic processes in the body. Reactome does not classify as drugs those substances produced naturally by the human body (such as insulin or epinephrine), though these may be administered as medicines and may be considered drugs in other databases. Artificial derivatives of naturally produced human substances (such as glargine, a derivative of insulin) are considered drugs, as are non-human natural substances such as quinine and caffeine. All Drug subclass instances share three mandatory attributes: disease (the disease treated), referenceEntity (the Guide to Pharmacology cross-reference), and compartment (the subcellular location).

Drug *[superclass]*

A superclass that is not used to instantiate specific instances. The Drug superclass contains subclasses ChemicalDrug, ProteinDrug, and RNADrug. A drug may be defined as any substance administered as a medicine to alter metabolic processes in the body, most often processes involved in diseases or abnormal states.

ChemicalDrug

A subclass of Drug. A ChemicalDrug is a synthetic substance or non-human natural substance that is not a protein, RNA, or DNA molecule and is administered to the body to modify a metabolic reaction or reactions, usually to treat a disease or abnormal state.

Attribute / Relationship	Description
disease → Disease	One or more Disease Ontology terms identifying the disease(s) for which this drug is used. Mandatory.
referenceEntity → ReferenceTherapeutic	A cross-reference to the corresponding record in Guide to Pharmacology, providing the official drug name (INN), approval status, and pharmacological type. Mandatory for all Drug subclass instances.
compartment → Compartment	The subcellular location at which this drug is active or to which it localises. Mandatory.
crossReference → DatabaseIdentifier	Optional additional cross-references to external databases such as ChEBI, PubChem, or DrugBank.
literatureReference → LiteratureReference	Optional primary literature references supporting the annotation of this drug in Reactome.
cellType → CellType	An optional Cell Ontology term restricting this drug's action or localisation to a specific cell type.

ProteinDrug

A subclass of Drug. A ProteinDrug is a substance containing an active component that is a protein, whether modified or unmodified, and is administered to the body to modify a metabolic reaction or reactions, usually to treat a disease or abnormal state. Examples: semaglutide (a chemically modified form of GLP1) and etanercept (a fusion between the TNFalpha receptor and the Fc portion of IgG1). Naturally occurring human proteins such as insulin are EntityWithAccessionedSequence instances, not ProteinDrugs. Artificially modified versions of human proteins, such as glargine or lispro insulin, are considered ProteinDrugs.

Attribute / Relationship	Description
disease → Disease	One or more Disease Ontology terms identifying the disease(s) treated. Mandatory.
referenceEntity → ReferenceTherapeutic	The Guide to Pharmacology cross-reference. Mandatory for all Drug subclass instances.
compartment → Compartment	The subcellular location. Mandatory.
crossReference → DatabaseIdentifier	Optional additional cross-references.
literatureReference → LiteratureReference	Optional supporting literature references.

RNADrug

A subclass of Drug. An RNADrug is a substance containing an active component that is a ribonucleic acid, whether modified or unmodified, and is administered to the body to modify a metabolic reaction or reactions, usually to treat a disease or abnormal state. EntityWithAccessionedSequence is the non-drug class corresponding to RNADrug for naturally occurring human RNAs. Examples: inclisiran and asvasiran (non-natural small interfering RNAs, siRNAs).

Attribute / Relationship	Description
disease → Disease	The Disease Ontology term identifying the disease treated. Mandatory.
referenceEntity → ReferenceTherapeutic	The Guide to Pharmacology cross-reference. Mandatory for all Drug subclass instances.
compartment → Compartment	The subcellular location. Mandatory.
crossReference → DatabaseIdentifier	Optional additional cross-references.
literatureReference → LiteratureReference	Optional supporting literature references.

Section 4 — Modification Classes

Modification classes describe changes to biological sequences: post-translational modifications of proteins, genetic variants, and nucleotide modifications of RNAs. Attached to EntityWithAccessionedSequence instances via the hasModifiedResidue relationship.

AbstractModifiedResidue *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. The superclass from which GeneticallyModifiedResidue, TranscriptionalModification, and TranslationalModification inherit their shared attributes. Describes a modification to a reference sequence. Three subclass families: GeneticallyModifiedResidue (genetic variants), TranscriptionalModification (RNA modifications), TranslationalModification (protein PTMs).

4.1 Translational Modifications (Protein PTMs)

TranslationalModification *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Subclass of AbstractModifiedResidue. Describes a post-translational chemical modification to a protein. Modifications are cross-referenced to the PSI-MOD ontology. Three concrete subclasses: ModifiedResidue, GroupModifiedResidue, CrosslinkedResidue.

ModifiedResidue

Subclass of TranslationalModification. Used to describe the translational or post-translational modification of an amino acid where the chemistry can be specified in atomic detail using a PSI-MOD term — for example, phosphorylation, acetylation, gamma-carboxylation. ModifiedResidue is used when the

chemical nature of the modifying group can be fully specified by a psiMod term; GroupModifiedResidue is used when the modification cannot be completely described by a psiMod term.

Attribute / Relationship	Description
coordinate	The amino acid residue position in the reference sequence at which this modification occurs. Use UniProt residue numbering when available.
label	A short display label for this modification (e.g., 'phospho', 'acetyl'), used to build the EWAS display name.
psiMod → PsiMod	The PSI-MOD term describing the chemical nature of the modification. Required.
referenceSequence → ReferenceGeneProduct or ReferenceIsoform	The canonical protein sequence containing the modified residue. Required. Use ReferenceGeneProduct for the canonical sequence; ReferenceIsoform only when the modification is demonstrated specifically in a non-canonical isoform.

GroupModifiedResidue

Subclass of TranslationalModification. Describes the modification of an amino acid residue with a chemical entity that cannot be specified in atomic detail using psiMod alone — for example, attachment of a dextrin or glycogen moiety to a tyrosine side chain in glycogenin. Such incompletely specified chemical entities are beyond the scope of the PSI-MOD ontology but are available in ChEBI. The psiMod attribute describes the linkage type; the modification attribute (a ReferenceGroup or Polymer) describes the attached group. Example: limit dextrins on L-Tyrosine [ChEBI:17895] at position 194 of UniProt:P46976 GYG1 — psiMod is MOD:00166, modification is CHEBI:28912.

Attribute / Relationship	Description
psiMod → PsiMod	The PSI-MOD term describing the chemical linkage between the residue and the attached group. Required.
referenceSequence → ReferenceGeneProduct or ReferenceIsoform	The parent protein sequence. Required.
modification → ReferenceGroup, Polymer, or DefinedSet	The chemical group, polymer moiety, or set of chemical entities attached to the residue.

CrosslinkedResidue *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Subclass of TranslationalModification. Two subclasses — InterChainCrosslinkedResidue and IntraChainCrosslinkedResidue — annotate disulfide bonds, thioester bonds, isopeptide bonds, and other crosslinks including to sumoyl or ubiquitin groups.

IntraChainCrosslinkedResidue

Subclass of CrosslinkedResidue. Used to annotate cross-linking between amino acid residues within the same peptide chain — for example, a disulfide bond within a single protein. IntraChainCrosslinkedResidue instances are made in pairs, one associated with each cross-linked residue in the EWAS record, and the two records are linked via their equivalentTo attribute.

Attribute / Relationship	Description
coordinate	The position of the first cross-linked residue.

secondCoordinate	The position of the second cross-linked residue in the same chain.
psiMod → PsiMod	The PSI-MOD term describing the type of crosslink (e.g., disulfide bond). Required.
referenceSequence → ReferenceGeneProduct	The protein chain containing both cross-linked residues. Required.
modification → ReferenceGroup	The chemical group forming the crosslink (e.g., the sumoyl or ubiquitin group in an isopeptide bond), when applicable.
equivalentTo → IntraChainCrosslinkedResidue	Points to the partner IntraChainCrosslinkedResidue record for the other participating residue. This pair linkage is mandatory.

InterChainCrosslinkedResidue

Subclass of CrosslinkedResidue. Used to annotate cross-linking between residues on two different peptide chains — for example, an inter-chain disulfide bond in an antibody or an isopeptide bond between ubiquitin and a substrate lysine. A pair of InterChainCrosslinkedResidue records is created, one on each EWAS involved in the crosslink, linked via equivalentTo. The secondReferenceSequence attribute identifies the partner chain.

Attribute / Relationship	Description
coordinate	The position of the cross-linked residue in the primary chain.
secondCoordinate	The position of the cross-linked residue in the partner chain.
psiMod → PsiMod	The PSI-MOD term for the crosslink type. Required.
referenceSequence → ReferenceGeneProduct	The primary protein chain. Required.
secondReferenceSequence → ReferenceGeneProduct or ReferenceDNASequence	The partner protein or DNA chain. Required.
modification → ReferenceGroup	The chemical group forming the crosslink, where applicable.
equivalentTo → InterChainCrosslinkedResidue	Points to the partner InterChainCrosslinkedResidue on the other EWAS. Required.

4.2 Transcriptional Modifications

TranscriptionalModification *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Subclass of AbstractModifiedResidue. Describes an RNA sequence modification.

ModifiedNucleotide

Describes the location and chemical nature of a nucleotide modification in an RNA sequence — for example, pseudouridine or N6-methyladenosine (m6A). Attached to RNA EWAS instances via hasModifiedResidue.

Attribute / Relationship	Description
coordinate	The nucleotide position in the reference RNA sequence at which this modification occurs.
referenceSequence → ReferenceRNASequence or ReferenceDNASequence	The parent RNA or DNA sequence. Required.
modification → ReferenceGroup, Polymer, or ReferenceMolecule	The chemical entity constituting the modification.

4.3 Genetic Modifications

Describe sequence changes in variant (disease or engineered) gene products relative to the canonical UniProt reference sequence. All are subclasses of GeneticallyModifiedResidue and attached to variant EWAS instances via hasModifiedResidue.

GeneticallyModifiedResidue *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Subclass of AbstractModifiedResidue from which the FragmentModification and ReplacedResidue classes inherit shared attributes. Describes a protein sequence change resulting from a genetic modification.

FragmentModification *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Children of this class (FragmentDeletionModification, FragmentInsertionModification, and FragmentReplacedModification) capture the deletion, insertion, or replacement of consecutive amino acid residues in an EWAS in a disease variant of a gene product.

Attribute / Relationship	Description
startPositionInReferenceSequence	The first amino acid position of the affected segment in the reference sequence.
endPositionInReferenceSequence	The last amino acid position of the affected segment in the reference sequence.
referenceSequence → ReferenceGeneProduct or ReferenceIsoform	The canonical protein sequence in which this fragment-level modification occurs. Required.

FragmentDeletionModification

Records the deletion of a continuous fragment of an EntityWithAccessionedSequence. Example: deletion of amino acid residues 30 to 297 in the EGFR mutant protein EGFRvIII in glioblastoma. Attributes are the referenceSequence (UniProt P00533, EGFR), the startPositionInReferenceSequence (30 for EGFRvIII), and the endPositionInReferenceSequence (297 for EGFRvIII).

FragmentInsertionModification

Used to describe modification of an EWAS through insertion of a continuous fragment — used for duplications and fusions. For duplications, the referenceSequence is the same as the EWAS referenceEntity (the inserted sequence is a direct repeat of adjacent amino acids). For fusions, the referenceSequence in FragmentInsertionModification is the referenceEntity for the C-terminal fusion partner, differing from the N-terminal partner specified in the EWAS record.

Attribute / Relationship	Description
coordinate	The junction position at which the insertion occurs.
startPositionInReferenceSequence	The start of the inserted fragment in the C-terminal partner's reference sequence (for fusions).
endPositionInReferenceSequence	The end of the inserted fragment in the reference sequence.
referenceSequence → ReferenceGeneProduct or ReferenceIsoform	For duplications: same as EWAS referenceEntity. For fusions: the C-terminal partner's reference sequence. Required.

FragmentReplacedModification

Records replacement of the WT amino acid sequence with a specified alternate sequence, arising from an insertion, fusion, indel, frameshift, or extension. Unlike the other FragmentModification subclasses, this class requires that the curator explicitly annotate the single-letter amino acid sequence of the variant region (alteredAminoAcidFragment). By Reactome convention, insertions are captured by FragmentReplacedModification, not FragmentInsertionModification. For an insertion, the alteredAminoAcidFragment includes both the flanking WT 5'- and 3' amino acids AND the inserted amino acids. Example: for EGFR D770_N771insNPG, the alteredAminoAcidFragment is DNPGN.

Attribute / Relationship	Description
alteredAminoAcidFragment	The single-letter amino acid sequence of the variant region, including flanking WT residues for insertions.
startPositionInReferenceSequence	The start of the affected region in the WT reference sequence.
endPositionInReferenceSequence	The end of the affected region in the WT reference sequence.
referenceSequence → ReferenceGeneProduct or ReferenceIsoform	The WT reference sequence. Required.

ReplacedResidue

A subclass of GeneticallyModifiedResidue. Describes the replacement of an amino acid residue with a different residue or a stop codon as the result of a mutation (germline or somatic). Class attributes identify the coordinate of the affected residue, the identity of the residue in the UniProt sequence (first psiMOD value), and the identity of the replacement residue (second psiMOD value).

Attribute / Relationship	Description
coordinate	The position in the reference sequence of the substituted residue.
psiMod → PsiMod	Two PSI-MOD values: (1) the WT amino acid at this position; (2) the replacement amino acid. For NonsenseMutation, only one psiMod is provided.

referenceSequence → ReferenceGeneProduct or ReferenceIsoform	The parent protein sequence. Required.

NonsenseMutation

A subclass of the ReplacedResidue class. Describes the coordinate at which a protein referenceSequence is truncated due to a nonsense mutation. In the NonsenseMutation record, a single psiMod is specified, indicating as an amino acid removal the amino acid that is altered to generate a stop codon.

Section 5 — Catalysis and Regulation Classes

CatalystActivity

Describes the catalyst in a ReactionLikeEvent: associates one physicalEntity with one GO_MolecularFunction. Example: cysteine-type endopeptidase activity of active caspase-8. If the physicalEntity is a Complex and a component of the complex mediates the molecularFunction, that component should be identified as the activeUnit of the CatalystActivity. If a PhysicalEntity can enable multiple MolecularFunctions in the same reaction context, a separate CatalystActivity instance is created for each.

Attribute / Relationship	Description
activity → GO_MolecularFunction	The GO Molecular Function term describing the catalytic activity. Use the most specific applicable GO MF term. Verified via OLS before assignment. Required.
physicalEntity → any PhysicalEntity	The PhysicalEntity performing the catalytic function (EWAS, Complex, DefinedSet, CandidateSet, etc.). Required.
activeUnit → any PhysicalEntity	The specific subunit of a Complex bearing the catalytic activity, when known and supported by literature. Leave blank if the whole complex acts, if the active subunit is unknown, or if the physicalEntity is not a Complex. The physicalEntity itself must never be its own activeUnit.

CatalystActivityReference

Associates a CatalystActivity instance with one or more Publications providing direct experimental evidence for that specific catalytic activity. Distinct from the literatureReference on the RLE: CatalystActivityReference provides evidence specifically for the catalytic mechanism. Linked from the ReactionLikeEvent via the catalystActivityReference relationship.

Attribute / Relationship	Description
catalystActivity → CatalystActivity	The specific CatalystActivity instance whose evidence is being recorded. Required.
literatureReference → LiteratureReference or Book	The publication(s) providing direct experimental evidence for this catalytic activity.

Regulation *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Consists of subclasses NegativeRegulation and PositiveRegulation, along with their subclasses NegativeGeneExpressionRegulation, PositiveGeneExpressionRegulation, and Requirement. Instances of Regulation subclasses are added to the regulatedBy attribute of ReactionLikeEvents to capture the negative or positive effect of a PhysicalEntity on the described event.

PositiveRegulation

An optional class used to populate the regulatedBy attribute of a ReactionLikeEvent when describing a stimulatory effect of a Regulator (a PhysicalEntity) on that event. The mechanism of action of the Regulator, if known, can be described through the optional activity attribute populated by GO_MolecularFunction terms. If the Regulator is a Complex, the specific Complex component(s) playing the regulatory role can be specified as activeUnit(s).

Attribute / Relationship	Description
regulator → any PhysicalEntity	The PhysicalEntity exerting the stimulatory effect. Required.
activity → GO_MolecularFunction	Optional. The GO Molecular Function term describing the mechanism of action, when known.
activeUnit → any PhysicalEntity	Optional. The specific subunit of a Complex regulator that directly mediates the regulatory activity.

NegativeRegulation

An optional class used to populate the regulatedBy attribute of a ReactionLikeEvent when describing an inhibitory effect of a Regulator on that event. The mechanism of action, if known, can be described through the optional activity attribute. If the Regulator is a Complex, the specific inhibitory component(s) can be specified as activeUnit(s).

Attribute / Relationship	Description
regulator → any PhysicalEntity	The PhysicalEntity exerting the inhibitory effect. Required.
activity → GO_MolecularFunction	Optional. The molecular function through which the regulator inhibits.
activeUnit → any PhysicalEntity	Optional. The active inhibitory subunit of a Complex regulator.

PositiveGeneExpressionRegulation

A subclass of PositiveRegulation describing a direct stimulatory effect of a Regulator on a BlackBoxEvent representing gene expression. The Regulator represents a complex of regulatory component(s) such as transcription factors with the target gene or mRNA molecule. The regulatory component(s) are defined as activeUnit(s). Used as part of the gene expression regulation annotation pattern: TF:gene complex → BBE → PositiveGeneExpressionRegulation.

Attribute / Relationship	Description
regulator → Complex or EWAS	For Pattern A (direct binding demonstrated): the TF:gene complex. For Pattern B (effect without demonstrated direct binding): the TF directly. Required.

activeUnit → any PhysicalEntity	The transcription factor or regulatory component(s) mediating activation. Required when regulator is a complex.
activity → GO_MolecularFunction	Optional. The DNA-binding or transcription activation activity.

NegativeGeneExpressionRegulation

A subclass of NegativeRegulation describing a direct inhibitory effect of a Regulator on a BlackBoxEvent representing gene expression or translation. The Regulator is a complex of regulatory component(s) with the target gene or mRNA. Also used for miRNA-mediated translational repression, where the Regulator is a RISC:miRNA:mRNA complex and the activeUnit is the RISC component.

Attribute / Relationship	Description
regulator → Complex or EWAS	The repressor complex or direct repressor. Required.
activeUnit → any PhysicalEntity	The repressor subunit(s) of a Complex regulator. Required when regulator is a complex.
activity → GO_MolecularFunction	Optional. The repressor's molecular function.

Requirement

A subclass of PositiveRegulation denoting a regulatory PhysicalEntity without which the regulated event cannot occur — a necessary enabling condition rather than a catalytic or modulatory factor. Use only for non-trivial requirements: do not create a Requirement for an entity already listed as an input.

Attribute / Relationship	Description
regulator → any PhysicalEntity	The PhysicalEntity required for the event to occur. Required.
activity → GO_MolecularFunction	Optional. The molecular function through which this required entity enables the event.
activeUnit → any PhysicalEntity	Optional. The active subunit, if the regulator is a complex.

RegulationReference

Associates any kind of Regulation instance with one or more Publications providing experimental evidence for that regulatory relationship. Linked from the ReactionLikeEvent via the regulationReference relationship and from the Regulation instance itself.

Attribute / Relationship	Description
literatureReference → LiteratureReference	The publication(s) providing experimental evidence for this regulatory relationship. Required.
regulation → any Regulation subclass	The Regulation instance whose evidence is being recorded.

ControlReference *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Associates one or more Publications with an instance of another class — specifically a CatalystActivity, Regulation, or Cell marker. Three concrete subclasses: CatalystActivityReference, RegulationReference, MarkerReference.

MarkerReference

Associates an RNA or protein EntityWithAccessionedSequence that has been identified as a proteinMarker or RNAMarker of a Cell with one or more Publications. MarkerReference is an attribute of the Cell class of PhysicalEntity, providing literature evidence for the marker assignments.

Attribute / Relationship	Description
cell → Cell	The Cell instance whose marker is being evidenced. Required.
marker → EWAS	The protein or RNA marker EWAS instance. Required.
literatureReference → LiteratureReference	The publication(s) supporting use of this EWAS as a marker for this Cell type. Required.

Section 6 — Disease Annotation Classes

EntityFunctionalStatus

EntityFunctionalStatus is used to mark a disease ReactionLikeEvent as either a gain- or a loss-of-function, and to describe the underlying reason for the change in behaviour. It has three attributes: diseaseEntity (the variant EWAS, Complex, or set of variants responsible for the phenotype), normalEntity (the wild-type instance corresponding to the disease PhysicalEntity), and functionalStatus (which relates the molecular change to the functional consequence). Required on all FailedReaction instances.

Attribute / Relationship	Description
diseaseEntity → Complex, DefinedSet, CandidateSet, or EWAS	The variant PhysicalEntity responsible for the disease phenotype. Required.
normalEntity → Complex or EWAS	The wild-type PhysicalEntity in the corresponding normal reaction. Required.
functionalStatus → FunctionalStatus	One or more FunctionalStatus instances combining a FunctionalStatusType term with a SequenceOntology structural variant term to characterise the molecular basis of the functional change. Required.

FunctionalStatus

Used to describe the molecular basis of a change in function. Combines a term from the FunctionalStatusType controlled vocabulary (identifying the alteration in gene function, e.g., gain-of-function) with a structuralVariant term from the SequenceOntology ExternalOntology (describing the nature of the genetic change, e.g., copy number gain). FunctionalStatus is used to populate the functionalStatus attribute of EntityFunctionalStatus.

Attribute / Relationship	Description

functionalStatusType → FunctionalStatusType	The controlled vocabulary term describing the functional consequence of this genetic variant. Required.
structuralVariant → SequenceOntology	A Sequence Ontology term describing the nature of the underlying genetic change — for example, missense variant (SO:0001583), frameshift variant (SO:0001589), copy number gain (SO:0001742). Required.

FunctionalStatusType

A subclass of ControlledVocabulary providing the controlled list of terms describing the functional consequence of a variant relative to its wild-type counterpart. Populates the functionalStatusType attribute of FunctionalStatus. Vocabulary maintained by the Editor-in-Chief. Current terms:

Attribute / Relationship	Description
loss_of_function	The variant eliminates or abrogates the normal molecular function of the protein.
gain_of_function	The variant confers a new or enhanced molecular function not present in the wild-type protein.
partial_loss_of_function	The variant reduces but does not completely eliminate the normal molecular function.
decreased_transcript_level	The variant reduces the steady-state abundance of the transcript, rather than directly affecting protein function.
increased_translational_product_level	The variant increases the steady-state abundance of the protein.

Section 7 — Reference Entity Classes

ReferenceEntity captures the invariant features of a molecule — its names, molecular structure, and links to external community reference databases such as UniProt, ENSEMBL, or ChEBI — and is created by importing content from those databases. A ReferenceEntity is an attribute of the PhysicalEntity subclasses Drug, EWAS, and SimpleEntity, and enables grouping of Reactome PhysicalEntities that represent the same molecule in different compartments or modification states. For example, 'Glc [extracellular region]' and 'Glc [cytosol]' are both linked to the same ReferenceMolecule for alpha-D-glucose, indicating their identical chemical nature. ReferenceEntities are not used in Reactions directly; they are attached to the PhysicalEntities involved. ReferenceEntities derived from UniProt and ChEBI are usually imported automatically. Contact the Managing Editor for missing ChEBI entries.

ReferenceEntity *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. The root superclass for all reference records. Shared properties: identifier (the accession in the external database), databaseName (the source database name), name, url (link to the source record).

ReferenceSequence *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Molecules with an accessioned sequence. A subclass of ReferenceEntity and superclass of ReferenceDNASequence, ReferenceGeneProduct, ReferenceIsoform, and ReferenceRNASequence. Instances should not be created manually. Additional shared properties: sequenceLength, geneName (HGNC gene symbol), otherIdentifier, secondaryIdentifier, description, keyword, checksum, isSequenceChanged (set to true

when the source database sequence has been updated, flagging that Reactome coordinate-dependent annotations may need review).

ReferenceGeneProduct

Used to populate the referenceEntity attribute for EWAS instances describing proteins. The canonical form of the protein from UniProt is used as the ReferenceGeneProduct unless experimental literature specifically supports the role of a particular isoform in the described function, in which case ReferenceIsoform is used. ReferenceGeneProduct is also used to populate the referenceSequence attribute of ModifiedResidue and related modification classes.

Attribute / Relationship	Description
identifier	The UniProt accession (e.g., P00533 for EGFR).
databaseName	The source database, typically 'UniProt'.
geneName	The HGNC gene symbol(s), imported from UniProt.
sequenceLength	The protein length in amino acids, imported from UniProt.
checksum	A sequence checksum used to detect when the UniProt sequence has been updated.
isSequenceChanged	Boolean. True when the UniProt source sequence has been updated since Reactome last imported it, indicating that coordinate-dependent annotations should be reviewed.
referenceDatabase → ReferenceDatabase	Points to the ReferenceDatabase instance for UniProt.
species → Species	The species of this protein.
referenceGene → ReferenceDNASequence	Links to the corresponding ENSEMBL gene record(s).
referenceTranscript → ReferenceRNASequence	Links to the corresponding ENSEMBL transcript(s).
crossReference → DatabaseIdentifier	Cross-references to other databases such as RefSeq, PDB, and Ensembl Protein.

ReferenceIsoform

Used to populate the referenceEntity attribute for EWAS instances describing proteins where experimental literature specifically supports the role of a defined non-canonical isoform. Use ReferenceGeneProduct for the canonical UniProt sequence; use ReferenceIsoform only when justified by specific isoform-level evidence in the literature.

Attribute / Relationship	Description
identifier	The UniProt isoform accession including the isoform suffix (e.g., P00533-2).
variantIdentifier	The isoform-specific variant identifier within the UniProt record.
databaseName	The source database, typically 'UniProt'.
isoformParent → ReferenceGeneProduct	Links this isoform record back to the canonical ReferenceGeneProduct from which it derives.
referenceDatabase → ReferenceDatabase	Points to the UniProt ReferenceDatabase instance.

species → Species	The species of this isoform.

ReferenceDNASequence

Used to populate the referenceEntity attribute for EWAS instances describing DNA molecules (for instance, EWAS instances of human genes in gene expression annotations). Also linked from ReferenceGeneProduct via the referenceGene relationship. The record consists of a referenceDatabase (generally ENSEMBL), an ENSEMBL gene identifier, and a species.

ReferenceRNASequence

Used to populate the referenceEntity attribute for EWAS instances describing RNA molecules (for instance, EWAS instances of mRNAs) and to populate the referenceSequence attribute of ModifiedNucleotide records. Also linked from ReferenceGeneProduct via the referenceTranscript relationship. The record consists of a referenceDatabase (generally ENSEMBL), an ENSEMBL transcript identifier, and optionally a species.

ReferenceMolecule

Used to cross-reference individual chemical molecules to the ChEBI database. Populates the referenceEntity attribute of SimpleEntity instances. The record consists of a ChEBI accession identifier, a referenceDatabase, a chemical formula, and a name.

ReferenceGroup

A chemical group instance (children of CHEBI:24433 — chemical group) from the ChEBI database. Used to populate the modification attribute of GroupModifiedResidue, InterChainCrosslinkedResidue, IntraChainCrosslinkedResidue, and ModifiedNucleotide instances. The record consists of a ChEBI accession identifier and a referenceDatabase.

ReferenceTherapeutic

Used to populate the referenceEntity attribute for PhysicalEntities of the Drug class, including ChemicalDrug, ProteinDrug, and RNADrug. The record provides the official drug identity from Guide to Pharmacology, including the International Non-proprietary Name (INN), approval status, and pharmacological type.

Attribute / Relationship	Description
identifier	The Guide to Pharmacology accession for this drug.
databaseName	Always 'Guide to Pharmacology'.
inn	The International Non-proprietary Name (INN) of the drug — the official non-commercial name adopted by the WHO.
approved	Boolean. Whether this drug has received regulatory approval (e.g., from the FDA or EMA). Imported from Guide to Pharmacology.
approvalSource	The regulatory body or source of the approval status information.

type	The pharmacological type according to Guide to Pharmacology: Synthetic organic, Peptide, Natural product, Antibody, Metabolite, etc.
referenceDatabase → ReferenceDatabase	Points to the Guide to Pharmacology ReferenceDatabase instance.
crossReference → DatabaseIdentifier	Cross-references to ChEBI, PubChem, DrugBank, and other external drug databases.

ReferenceDatabase

This class describes the source (external database) of an identifier in a DatabaseIdentifier or ReferenceEntity instance. For example, DatabaseIdentifier COMPOUND:C00113 links a Reactome instance to the record maintained by the KEGG 'compound' database — ReferenceDatabase describes KEGG and its URL templates.

Attribute / Relationship	Description
name	The canonical name of the database (e.g., 'UniProt', 'ChEBI', 'ENSEMBL', 'Guide to Pharmacology', 'KEGG COMPOUND').
url	The base URL of the database's web interface.
accessUrl	A URL template for constructing direct links to individual records (e.g., 'https://www.uniprot.org/uniprot/###ID###'). Used by the Reactome website to generate hyperlinks. Do not modify; contact developers if changes appear necessary.
resourceIdentifier	An optional identifiers.org-registered resource identifier for QA purposes. Does not affect routine curation.

DatabaseIdentifier

Refers to an external database identifier. An instance of this class is used for the crossReference attribute of Reactome instances to associate them with their representation in external ReferenceDatabases, in addition to the required referenceEntity cross-references. Example: the Reactome SimpleEntity instance PQQ (pyrroloquinoline quinone), cross-referenced to ReferenceMolecule ChEBI:18315, has a crossReference DatabaseIdentifier COMPOUND:C00113 linking it to the KEGG 'compound' database record.

Attribute / Relationship	Description
identifier	The accession in the external database.
databaseName	The name of the external database (automatically denormalized from the associated ReferenceDatabase instance).
url	The direct URL to this specific record in the external database, constructed from the ReferenceDatabase accessUrl template.

Section 8 — External Ontology Classes

An ExternalOntology is one developed and maintained outside of Reactome and used as-is to align Reactome annotation with community vocabularies. All ExternalOntology subclasses carry the following shared properties: identifier (the accession in the external ontology), databaseName, name, definition, synonym, url. This class should not be used for instance creation.

ExternalOntology *[superclass]*

Root superclass for all external ontology term classes. Not used for manual annotation or direct instance creation.

GO_Term *[superclass — graph co-label]*

Gene Ontology term used by Reactome events as cross-references. Can be GO_BiologicalProcess, GO_CellularComponent, or GO_MolecularFunction. In the graph database, the label GO_Term is co-applied to all three GO term class nodes as a common supertype, allowing queries of the form MATCH (n:GO_Term) to retrieve all GO terms regardless of category. The accession property holds only the numeric portion of the GO ID (e.g., '0005737' for cytosol); the full identifier must be constructed as 'GO:' + n.accession.

GO_BiologicalProcess

A local copy of the Gene Ontology Biological Process ontology. Instances provide the goBiologicalProcess attribute for Reactome Events (primarily Pathway instances). When adding a new pathway, the goBiologicalProcess attribute should be set to the most specific applicable GO BP term, verified as current and not obsolete via OLS. If no specific term exists for the process being annotated, follow the GO Term Gap annotation procedure in the Curator Guide.

GO_CellularComponent

A local copy of the Gene Ontology Cellular Component ontology. Instances provide the subcellular location of Reactome Events and PhysicalEntities. A curated subset of GO CC terms is designated as the Compartment vocabulary (see Compartment entry). The GO CC ontology hierarchy is navigable via componentOf (parent term links), surroundedBy (spatial containment), instanceOf (is_a relationships), and hasPart (part_of relationships).

Compartment

A subset of GO_CellularComponent terms manually curated by Reactome to specify non-overlapping subcellular locations as values for the compartment attribute of Events and PhysicalEntities. If a PhysicalEntity occurs in more than one compartment, different instances of the entity are created for each location, distinguished by their compartment attributes. Curatorial rules: use 'nucleoplasm' not 'nucleus'; use 'cytosol' not 'cytoplasm'; use the most specific applicable Compartment term. For transmembrane complexes spanning multiple compartments, specify the primary compartment (typically the membrane) and allow the includedLocation attribute to be auto-populated.

GO_MolecularFunction

A local copy of the Gene Ontology Molecular Function ontology. Instances provide the activity attribute of CatalystActivity instances and the optional activity attribute of Regulation instances. The most specific available GO MF term should be used, verified via OLS. Curators should never create GO_MolecularFunction instances directly — if a required term is not in Reactome, contact the Managing Editor. For enzyme activities, an EC number may be available in the ecNumber property.

Attribute / Relationship	Description
accession	The numeric portion of the GO accession (e.g., '0004672' for protein kinase activity). The full identifier is 'GO:' + accession.
name	The official GO term name.
definition	The official GO term definition, imported from GO.
ecNumber	The EC (Enzyme Commission) number associated with this enzyme activity, when one exists.

Anatomy

Anatomy encompasses terms representing body parts, organs, and tissues from the ExternalOntology UBERON (https://obophenotype.github.io/uberon/). Anatomy terms are used to populate the organ, tissue, and tissueLayer attributes of the Cell subclass of PhysicalEntity, and the tissue attribute of CellDevelopmentStep and CellLineagePath subclasses of Event.

CellType

CellType terms represent different categories of animal cells, characterised by their specific structure, function, and biochemistry, derived from the ExternalOntology Cell Ontology (https://obofoundry.org/ontology/cl.html). CellType terms are used to populate the cellType attribute of Cell and EntityWithAccessionedSequence (and other PhysicalEntity) subclasses.

Disease

Disease encompasses terms describing impairments of normal functions caused by pathogenic genetic mutations or environmental factors such as microbial pathogens. Terms are derived from the Disease Ontology (DO) (https://disease-ontology.org/) and are used to populate the disease attribute of abnormal PhysicalEntities (GenomeEncodedEntity, Complex, EntitySet) and abnormal events (ReactionLikeEvent, Pathway). Disease instances include identifier and name attributes taken from DO.

PsiMod

PsiMod encompasses terms representing post-translational chemical modifications of proteins, derived from the ExternalOntology Protein Modification Ontology (MOD: https://ontobee.org/ontology/MOD). The psiMod attribute is present on all subclasses of TranslationalModification (ModifiedResidue, GroupModifiedResidue, CrosslinkedResidue) and on the genetic modification classes ReplacedResidue and NonsenseMutation. The label property provides a short display descriptor unique to PsiMod among ExternalOntology classes.

SequenceOntology

SequenceOntology is a local copy of portions of the ExternalOntology Sequence Ontology (http://www.sequenceontology.org/). It encompasses terms representing the variant features and attributes of biological sequences. SequenceOntology terms are used to populate the structuralVariant attribute of FunctionalStatus instances, describing the nature of the genetic change giving rise to a disease variant (e.g., missense variant, frameshift variant, copy number gain).

EvidenceType

EvidenceType is a subclass of ExternalOntology, populated with Gene Ontology evidence code terms. Used only to flag inferred orthologous events created computationally as part of the release process (IEA, inferred by electronic annotation) and not for human manual annotation by curators, with one exception described below.

Attribute / Relationship	Description
IEA — inferred by electronic annotation	Set automatically on all computationally inferred orthologous events. Not set manually by curators.
biological system reconstruction evidence based on inference from background scientific knowledge	The only code that curators may set manually. Used in viral infection pathways where a reaction is annotated as occurring in a canonical human host cell infected by a canonical virus strain, but the supporting experimental evidence derives from a different viral strain or host species. When this code is used, the summation must explicitly state the actual evidence strain/species and explain the basis for the inference.
EXP — experimental evidence used in manual assertion	Present in the database as a legacy data entry on one reaction (R-HSA-9764791). Not an active curation practice; should not be set by curators.

> ⚠ *EvidenceType should not be used as a general mechanism for flagging weaker evidence. The only legitimate manual use is in the specific viral infection pathway context described above. Curators uncertain about evidence quality should consult the Managing Editor.*

Section 9 — Controlled Vocabulary Classes

ControlledVocabulary classes provide lists of permitted attribute values created and maintained internally by Reactome (as distinct from external ontology classes imported from community resources). Vocabularies are maintained by the Editor-in-Chief; curators should not add new terms without approval.

ControlledVocabulary *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. Lists of words that can be used in attributes in specific classes, optionally connected to one another with is_a relationships, created by Reactome as opposed to ones imported from external reference resources. Concrete subclasses: ReviewStatus, ReactionType, DrugActionType, DeletedControlledVocabulary, NegativePrecedingEventReason, and FunctionalStatusType.

ReactionType

Terms used as attributes to classify ReactionLikeEvents in ways that support data mining and classification — for example, in connection with the conversion of Reactome events into GO-CAM models. Currently used only to classify reactions involving drugs by mode of action.

> ⚠ *In the current database, ReactionType and DrugActionType are co-labels on the same three node instances. All three terms (Agonist, Antagonist, Inhibitor) carry both labels. This means that in practice these two classes are not separately populated — they represent two views of the same vocabulary. The data model team should verify whether this is intentional.*

DrugActionType

Terms used as attributes to classify ReactionLikeEvents involving drugs, to indicate the mode of action of the drug in the ReactionLikeEvent. Reactions that have drugs and sets or complexes containing drugs as inputs and output have a drugActionType attribute taken from the IUPHAR/Guide to Pharmacology glossary to classify the effect of the drug on the reaction (agonist, antagonist, inhibitor, etc.). See the ReactionType note above regarding co-labelling.

Section 10 — Publication Classes

The Publication superclass encompasses any document, paper, or electronic resource that makes information available to the public. Subclasses are Book, LiteratureReference, and URL. Publications are used in the literatureReference attribute of events, entities, Summation instances, and ControlReference subclass instances.

Publication *[superclass]*

Used to maintain logical integrity of the data model; not used for manual annotation. The Publication superclass encompasses any document, paper, or electronic resource that makes information available to the public. Subclasses: Book, LiteratureReference, and URL.

LiteratureReference

A journal article, typically abstracted in PubMed, cited in a summation or linked directly to an entity or event instance as primary evidence. The pubMedIdentifier must be supplied manually by the curator; once entered, the curator tool automatically retrieves and fills in the title, journal, year, volume, pages, and author fields from PubMed.

Attribute / Relationship	Description
pubMedIdentifier	The PubMed identifier (PMID) for this article. A number only, without any prefix. Mandatory. Must be entered manually by the curator.
title	The title of the journal article. Auto-populated from PubMed once the PMID is entered.
journal	The name of the journal. Auto-populated from PubMed.
year	The year the article was published. Auto-populated from PubMed.
volume	The journal volume. Auto-populated from PubMed.
pages	The inclusive page numbers (e.g., '123-145'). Auto-populated from PubMed.
author → Person	The author(s) of this article, each as a Person instance. Auto-populated from PubMed.

Book

A book or book chapter used in the literatureReference attribute of relevant data class instances. Author, title, and year of publication are required. Chapters of books can also be cited using the chapterTitle attribute of the Book class.

Attribute / Relationship	Description
title	The title of the book.

year	The year of publication.
pages	The page range, for book chapters.
ISBN	The ISBN of the book.
author → Person	The author(s) of the book or chapter.
publisher → Affiliation	The publishing institution.

URL

A website cited as a reference in data class instances such as summations, entities, or events. The URL should contain all information, including transfer protocols such as 'http://', 'https://', or 'ftp://', required to navigate to the resource.

Attribute / Relationship	Description
uniformResourceLocator	The full URL including protocol. Required.
title	A descriptive title for this resource.
author → Person	Optional. The author(s) attributed to this resource.

Section 11 — Persons and Affiliations

Person

The name of a person. Used to identify curators, authors, reviewers, and literature reference authors. Any Person instance can have an affiliation attribute, but at present this information is recorded only for Reactome contributors (authors, reviewers, and curators). A contributor identifier such as an ORCID can be associated with a Person instance via the crossReference attribute, allowing contributors to claim their Reactome contributions in ORCID.

Attribute / Relationship	Description
surname	The family name of the person. Mandatory. For consortia cited as authors, the consortium name is entered as the surname.
initial	The first-name initial. Mandatory. Enter a dash '-' if no first name is available. For consortia, enter '-' as the initial.
firstname	The first name of the person. Optional. Reactome uses only the initial on the website.
orcidId	The ORCID identifier for this person, when available.
project	The Reactome project this person is associated with.
affiliation → Affiliation	The institutional affiliation. Recorded for Reactome authors, reviewers, and curators only; not for literature reference authors.
crossReference → DatabaseIdentifier	An ORCID cross-reference enabling this person to claim their Reactome contributions.

Affiliation

The name and address of an institution — for example, Cold Spring Harbor Laboratory / 1 Bungtown Road, Cold Spring Harbor NY USA. Any Person instance can have an affiliation, but at present this is recorded only for Reactome authors, reviewers, and curators.

Attribute / Relationship	Description
name	The name of the institution.
address	The address of the institution.

Section 12 — Taxonomy Classes

Taxon

A controlled vocabulary populated manually by Reactome from the NCBI Taxonomy Database, providing terms to identify the species of events and genome-encoded physical entities. Taxonomy terms of higher rank than species, and their relationships, are maintained in Reactome to facilitate computational reasoning and data mining. Each Taxon instance includes the taxon's name as assigned by NCBI, its NCBI identifier (via a crossReference), and its NCBI-assigned superTaxon. This class is maintained by the Editor-in-Chief and Managing Editor.

Attribute / Relationship	Description
taxId	The NCBI Taxonomy identifier for this taxon.
name	The NCBI-assigned name(s) for this taxon.
abbreviation	A short code for this species used in constructing Stable Identifiers (e.g., 'HSA' for Homo sapiens, 'MMU' for Mus musculus).
superTaxon → Taxon	Points to the parent taxon in the NCBI taxonomy hierarchy. For example, the superTaxon for both Rattus and Mus is Murina; the superTaxon for Murina is Muridae.

Species

Gene products and events have species-level NCBI-derived taxon terms as attributes. Maintained by the Editor-in-Chief and Managing Editor. The species attribute on an event identifies the organism in which the reaction is occurring. For human Reactome curation this should almost always be Homo sapiens. In disease or host-pathogen contexts, the host species is listed in the species attribute and the pathogen in the relatedSpecies attribute. When creating a chimeric reaction for cross-species inference (involving entities from multiple species, usually observed in vitro), ALL species should be listed in the species attribute.

Section 13 — Interaction Classes

Interaction classes represent molecular interactions imported automatically into Reactome from external databases. They do not participate in the Reactome pathway hierarchy and are not created by manual curation. They are displayed as interaction overlays in the Reactome Pathway Browser.

Interaction *[superclass]*

Parent label for all kinds of molecular interactions brought into Reactome automatically from other databases, based on referenced evidence in a reviewed publication. Concrete subclass: UndirectedInteraction.

UndirectedInteraction

A kind of molecular interaction for which no information is available about the directionality of the interaction. Interactions of this type currently originate from IntAct. Each interaction links to the molecular participants via the interactor relationship.

Attribute / Relationship	Description
databaseName	The source database from which this interaction was imported (currently always 'IntAct').
accession	The accession identifier for this interaction in the source database.
pubmed	The PubMed identifier of the publication providing experimental evidence for this interaction.
score	The interaction confidence score assigned by the source database (e.g., IntAct MI-score).
url	A direct URL to this interaction record in the source database.
interactor → ReferenceGeneProduct, ReferenceIsoform, ReferenceMolecule, or ReferenceGroup	The molecular participants. Each interaction links to at least two interactor nodes, pointing to the ReferenceEntity records identifying the participating molecules.
referenceDatabase → ReferenceDatabase	Points to the ReferenceDatabase instance for the source interaction database.

Appendix A — Complete Relationship Type Inventory

All relationship types present in the Reactome graph database. Each relationship is described in plain English — what it means biologically or organisationally, and which classes participate. Relationships marked † are auto-generated during the release pipeline and must not be set manually by curators.

Attribute / Relationship	Description
RNAMarker Cell → EWAS	Identifies a specific RNA molecule as a molecular marker of a Cell type or state. Populated on Cell instances to record the RNA species that are specifically expressed or upregulated in that cell. Each RNAMarker link should have a corresponding MarkerReference providing the publication evidence.
activeUnit CatalystActivity or Regulation → PhysicalEntity	Designates the specific subunit of a Complex that directly bears the catalytic activity or regulatory function. Used when only one component of a multi-subunit complex is responsible for the annotated molecular function. If the whole complex acts as a unit, or if the active subunit is not known, this attribute is left blank. The physicalEntity of a CatalystActivity must never be its own activeUnit.
activity CatalystActivity or Regulation → GO_MolecularFunction	Specifies the GO Molecular Function term that describes the molecular activity performed by the catalytic entity or regulator. In CatalystActivity, this is the activity of the enzyme or other catalyst. In Regulation, this optionally describes the mechanism through which the regulator acts on the event.

affiliation Person → Affiliation	Links a Person to their institutional affiliation. Recorded only for Reactome contributors (curators, authors, reviewers) — not for the authors of literature references.
author Person → LiteratureReference, InstanceEdit, Book, or URL	Records the authorship of a publication or database edit. On LiteratureReference and Book instances, identifies the individual authors; on InstanceEdit, identifies the curator responsible for the database change.
authored † InstanceEdit → Event or PhysicalEntity	Records who was intellectually responsible for the biological content of an event or entity — that is, who wrote the biological annotation. Distinct from 'created', which records only the database commit. Used to populate the contributor information displayed on Reactome pathway and entity pages.
catalystActivity ReactionLikeEvent → CatalystActivity	Connects a reaction to its CatalystActivity record, which in turn specifies the catalytic entity and the GO Molecular Function it performs in that reaction.
catalystActivityReference ReactionLikeEvent → CatalystActivityReference	Connects a reaction directly to the publication(s) that provide experimental evidence specifically for the catalytic mechanism of that reaction, separately from the overall reaction evidence.
cell MarkerReference → Cell	Links a MarkerReference record back to the Cell instance for which the marker is being documented.
cellType PhysicalEntity → CellType	Restricts a PhysicalEntity to a specific cell type when its behaviour, modification state, or subcellular localisation is known to be cell-type-specific. Applied to EWAS, SimpleEntity, Complex, ChemicalDrug, and OtherEntity instances.
compartment Event or PhysicalEntity → Compartment	Specifies the subcellular location of a reaction, pathway, or physical entity. Must use terms from the Compartment vocabulary (a curated subset of GO CC terms). For entities, different compartments require separate instances of the same molecule.
componentOf † GO_CellularComponent → GO_CellularComponent	Represents the is_part_of (structural parent) relationship between GO Cellular Component terms, imported directly from the Gene Ontology. A child GO CC term is a componentOf its structural parent (e.g., nucleoplasm is componentOf nucleus).
created † InstanceEdit → any class	Records which database instance was created or first committed in a given InstanceEdit. Generated automatically at each database commit; spans all biological and infrastructure classes.
crossReference any class → DatabaseIdentifier	Provides a cross-reference to an external database record supplementary to the primary referenceEntity link. Used on PhysicalEntities, Events, ReferenceEntities, and Person instances to link to representations of the same object in other databases (e.g., KEGG COMPOUND, IntAct, ORCID).
deletedInstance Deleted → DeletedInstance	Connects a Deleted audit record to the companion DeletedInstance metadata record that preserves the class, stable identifier, display name, and other identifying information about the removed instance.
disease PhysicalEntity or Event → Disease	Associates an abnormal entity or event with the Disease Ontology term identifying the associated disease. Required on all FailedReaction instances and on disease variant entities.
diseaseEntity EntityFunctionalStatus → PhysicalEntity	Identifies the variant PhysicalEntity — the mutant protein, disease-associated complex, or set of variants — that is responsible for the disease phenotype described by this EntityFunctionalStatus record.
edited † InstanceEdit → Event	Records that an event's biological content was subsequently modified in this InstanceEdit, as distinct from its initial authorship.
entityFunctionalStatus FailedReaction → EntityFunctionalStatus	Links a FailedReaction to the EntityFunctionalStatus record that documents the variant entity and the molecular basis of its loss of function.

entityOnOtherCell Reaction or Complex → PhysicalEntity	Identifies a PhysicalEntity that is located on a neighbouring cell rather than on the cell being modelled. Used in cell-cell communication annotations — for example, a ligand on the surface of a signalling cell that is recognised by a receptor on the target cell.
equivalentTo CrosslinkedResidue → CrosslinkedResidue	Links the two members of a crosslinked-residue pair to each other. For IntraChainCrosslinkedResidue: each record (one per participating residue) points to the other. For InterChainCrosslinkedResidue: the record on each EWAS points to the partner record on the other EWAS.
evidenceType Event → EvidenceType	Associates an event with a GO evidence code. In practice, set automatically to IEA on all computationally inferred events. Curators may set it manually only in the specific viral inference context described in the EvidenceType class entry.
figure Event → Figure	Links an event to an illustrative figure image created by the Reactome graphic illustrator.
functionalStatus EntityFunctionalStatus → FunctionalStatus	Links an EntityFunctionalStatus to the FunctionalStatus record(s) that specify both the functional consequence (e.g., loss-of-function) and the nature of the underlying genetic change (e.g., missense variant).
functionalStatusType FunctionalStatus → FunctionalStatusType	Links a FunctionalStatus record to the controlled vocabulary term identifying the functional consequence of the variant gene product relative to its wild-type counterpart.
goBiologicalProcess Pathway or ReactionLikeEvent → GO_BiologicalProcess	Associates an event with the GO Biological Process term that best describes the biological process it represents. Primarily assigned to Pathway instances. Also set on some individual RLEs by curator decision in specific pathway families.
goCellularComponent PhysicalEntity → GO_CellularComponent	A supplementary compartment annotation using a GO CC term not in the curated Compartment vocabulary. Used primarily in infectious disease and host-pathogen pathways for 'host cell' location terms (e.g., host cell cytosol, host cell plasma membrane) that are not represented in the Compartment subset.
hasCandidate CandidateSet → PhysicalEntity	Identifies an unconfirmed or incompletely characterised candidate member of a CandidateSet — an entity thought to participate in the annotated reaction based on functional similarity or inference, but lacking direct experimental confirmation in this context.
hasComponent Complex → PhysicalEntity	Identifies each component of a Complex. Each component is entered individually; stoichiometry is indicated by a multiplicity prefix (2x, 3x) on the component name.
hasEncapsulatedEvent Pathway → Pathway	Identifies a pathway that is visually displayed within this pathway's diagram for contextual reference, but that is not a structural child in the Reactome event hierarchy.
hasEvent Pathway → ReactionLikeEvent or Pathway	Identifies the constituent events of a Pathway, entered in biological order. A Pathway can contain any mixture of RLE types (Reaction, BlackBoxEvent, FailedReaction, Polymerisation, Depolymerisation, CellDevelopmentStep) and sub-Pathways.
hasMember DefinedSet or CandidateSet → PhysicalEntity	Identifies a confirmed, functionally equivalent member of a DefinedSet, or a confirmed well-characterised member of a CandidateSet. All members must share the same compartment.
hasModifiedResidue EWAS → any AbstractModifiedResidue subclass	Links an EntityWithAccessionedSequence to each of its modification records — PTMs, genetic variants, nucleotide modifications. One record per modification.
hasPart † GO_CellularComponent → GO_CellularComponent	Represents the part_of relationship between GO Cellular Component terms, imported from the Gene Ontology (e.g., the nucleus hasPart the nucleoplasm).

includedLocation † Complex → Compartment	Auto-populated during the database release process. Records secondary compartments occupied by components of a transmembrane or multi-compartment complex. Curators specify only the primary compartment in the compartment attribute; this attribute is filled in automatically for the remaining compartments.
inferredTo † Event or PhysicalEntity → same class	Points from a curated human event or entity to the orthologous event or entity in another species that was computationally inferred from it during the Reactome cross-species inference pipeline.
input ReactionLikeEvent → PhysicalEntity	Identifies each PhysicalEntity consumed in this reaction, entered individually. Includes all stoichiometric participants: cofactors, cosubstrates, and small molecules such as ATP alongside primary substrates.
instanceOf † GO_CellularComponent → GO_CellularComponent	Represents the is_a relationship between GO Cellular Component terms, imported from the Gene Ontology.
internalReviewed † InstanceEdit → Event	Records that an event underwent internal Reactome review in this InstanceEdit.
interactor UndirectedInteraction → ReferenceEntity	Identifies a molecular participant in an interaction imported from an external interaction database such as IntAct. Each interaction links to at least two interactor nodes, each a ReferenceEntity record.
isoformParent ReferenceIsoform → ReferenceGeneProduct	Links a non-canonical isoform record back to the canonical ReferenceGeneProduct from which it derives, enabling grouping of all isoforms of the same gene product.
literatureReference Event, PhysicalEntity, Summation, or ControlReference → LiteratureReference	Links an event, entity, summation, or evidence record to a supporting publication. For events and entities, this provides the primary experimental evidence. For summations, it provides background supporting citations only.
marker MarkerReference → EWAS	Identifies the specific protein or RNA EWAS instance that is being used as a molecular marker for a Cell type in this MarkerReference record.
markerReference Cell → MarkerReference	Links a Cell instance to a MarkerReference record that documents the publication evidence for one of its molecular markers.
modification GroupModifiedResidue, InterChainCrosslinkedResidue, IntraChainCrosslinkedResidue, or ModifiedNucleotide → ReferenceGroup, Polymer, or DefinedSet	Identifies the chemical group, polymer, or set of chemical entities that constitute the modifying moiety attached to a protein or RNA residue. Used when the modifying group cannot be fully specified by a PSI-MOD term alone.
modified † DatabaseObject → InstanceEdit	Auto-populated. Records the InstanceEdit in which this instance was most recently modified.
negativePrecedingEvent ReactionLikeEvent → NegativePrecedingEvent	Links a reaction to a NegativePrecedingEvent record that flags a computationally inferred causal connection as having been manually reviewed and rejected by a curator.
normalEntity EntityFunctionalStatus → PhysicalEntity	Identifies the wild-type PhysicalEntity in the normal reaction, serving as the functional reference for the disease variant described in this EntityFunctionalStatus.
normalPathway Disease Pathway → Wild-type Pathway	Links a disease pathway variant to the corresponding wild-type pathway from which it was derived. The pathway-level counterpart to the normalReaction relationship on FailedReaction instances.

normalReaction FailedReaction → Reaction or BlackBoxEvent	Links a FailedReaction to the wild-type ReactionLikeEvent that proceeds normally in the absence of the disease variant.
organ Cell → Anatomy	Associates a Cell instance with the UBERON organ term identifying the organ in which this cell type resides.
output ReactionLikeEvent → PhysicalEntity	Identifies each PhysicalEntity produced in this reaction, entered individually. Includes all byproducts such as ADP, Pi, and other small molecules alongside primary products.
physicalEntity CatalystActivity → PhysicalEntity	Identifies the PhysicalEntity — EWAS, Complex, DefinedSet, CandidateSet, or other subclass — that performs the catalytic function described by this CatalystActivity instance.
precedingEvent ReactionLikeEvent or Pathway → ReactionLikeEvent or Pathway	Points to the event that immediately precedes this one in the biological sequence — typically the event whose output is consumed as input by the current event. When linking across pathway boundaries, always link to the specific terminal ReactionLikeEvent, not to the Pathway node.
previousReviewStatus Event → ReviewStatus	Records the review status of an event before its most recent change in review state, providing a one-step history of review progression.
proteinMarker Cell → EWAS	Identifies a specific protein as a molecular marker of a Cell type or state. Each proteinMarker link should have a corresponding MarkerReference providing the publication evidence.
psiMod ModifiedResidue, GroupModifiedResidue, CrosslinkedResidue, ReplacedResidue, or NonsenseMutation → PsiMod	Links a modification record to the PSI-MOD ontology term that describes the chemical nature of the modification — for example, the type of phosphorylation, the identity of amino acids in a crosslink, or the WT and replacement residues in a missense variant.
publisher Book → Affiliation	Links a Book publication to the institution that published it.
reactionType Reaction → ReactionType/DrugAction Type	Classifies the mode of drug action in a reaction that involves a Drug subclass PhysicalEntity as input or output. Terms are taken from the IUPHAR/Guide to Pharmacology glossary.
reason NegativePrecedingEvent → NegativePrecedingEventReason or Deleted → DeletedControlledVocabulary	Links a NegativePrecedingEvent to the term explaining why an inferred precedingEvent was rejected, or links a Deleted record to the term explaining why the instance was deleted.
referenceDatabase ReferenceEntity subclass or UndirectedInteraction → ReferenceDatabase	Links a reference entity or interaction record to the external database from which it was imported, providing access to the database URL and identifier template.
referenceEntity EWAS, SimpleEntity, or Drug subclass → ReferenceEntity subclass	The primary cross-reference linking a PhysicalEntity to its canonical record in an external reference database (UniProt, ENSEMBL, ChEBI, or Guide to Pharmacology). This is the definitive identity link for a PhysicalEntity, as distinct from crossReference which provides optional supplementary external links.
referenceGene ReferenceGeneProduct or ReferenceIsoform → ReferenceDNASequence	Links a protein reference record to the ENSEMBL gene record(s) encoding this protein, enabling navigation from protein to gene.

referenceSequence ModifiedResidue, GroupModifiedResidue, CrosslinkedResidue, FragmentModification, ReplacedResidue, NonsenseMutation, or ModifiedNucleotide → ReferenceSequence subclass	Links a modification record to the canonical reference sequence (protein or RNA) in which the modification occurs, providing the coordinate context for the modification.
referenceTranscript ReferenceGeneProduct or ReferenceIsoform → ReferenceRNASequence	Links a protein reference record to the ENSEMBL transcript(s) from which this protein is translated, enabling navigation from protein to transcript.
regulatedBy ReactionLikeEvent → any Regulation subclass	Links a reaction to a Regulation instance that describes a positive or negative regulatory effect on that reaction by a specific PhysicalEntity.
regulation RegulationReference → any Regulation subclass	Links a RegulationReference (the evidence record) back to the specific Regulation instance it provides experimental evidence for.
regulationReference ReactionLikeEvent → RegulationReference	Links a reaction directly to a RegulationReference providing publication evidence for a specific regulatory relationship of that reaction, separately from the overall reaction evidence.
regulator Regulation → PhysicalEntity	Identifies the PhysicalEntity that exerts the regulatory effect — stimulatory, inhibitory, or required — described by this Regulation instance.
relatedSpecies Event → Species	Identifies the 'bystander' species in a host-pathogen or symbiotic interaction event. The host species is listed in the species attribute; the pathogen or symbiont is listed here. Also used to list the pathogen species in infectious disease pathway annotations where the interaction is represented from the host's perspective.
release UpdateTracker → Release	Links a change record (UpdateTracker) to the specific Reactome release in which the recorded change was published.
repeatedUnit Polymer → PhysicalEntity	Identifies the building block(s) of a Polymer. A single value indicates a homo-polymer or fixed-stoichiometry polymer (where the repeating unit is itself a Complex of fixed composition). Multiple values indicate that the ratio of units is unknown or variable.
replacementInstances Deleted → any curated class	Points from a Deleted record to the replacement instance(s) in the live database, when the deleted instance was merged, split, or superseded rather than simply removed.
requiredInputComponent ReactionLikeEvent → PhysicalEntity	Identifies a PhysicalEntity that must be present for this event to occur but is not a stoichiometric reactant — for example, a specific protein domain or regulatory co-factor. Use only for non-trivial enabling conditions not already captured in the input list.
reverseReaction Reaction → Reaction	Points to the thermodynamically reversible counterpart of this reaction, when the reverse reaction has been curated as a separate Reaction instance.
reviewed † InstanceEdit → Event	Records that an event underwent external expert review in this InstanceEdit. Populates the reviewer information displayed on Reactome pathway pages.
revised † InstanceEdit → Event	Records that an event was revised following external expert review in this InstanceEdit.
reviewStatus Event → ReviewStatus	Associates an event with its current review status — a controlled vocabulary term indicating the level of internal and external review the event has undergone and whether it is included in public releases.

secondReferenceSequence InterChainCrosslinkedResidue → ReferenceGeneProduct or ReferenceDNASequence	Identifies the partner protein or DNA chain involved in an inter-chain crosslink — the chain to which the second cross-linked residue belongs.
species Event, PhysicalEntity, or ReferenceEntity → Species or Taxon	Identifies the biological species in which this event or entity is annotated. For chimeric reactions involving entities from multiple species, all involved species are listed.
structuralVariant FunctionalStatus → SequenceOntology	Links a FunctionalStatus record to the Sequence Ontology term describing the nature of the genetic change underlying the disease variant (e.g., missense variant, frameshift variant, copy number gain).
structureModified † InstanceEdit → Pathway	Records that the structural composition of a pathway — specifically its hasEvent membership — was changed in this InstanceEdit.
summation Event or PhysicalEntity → Summation	Links an event or entity to its free-text description. The summation may include background citations but should not contain the primary experimental evidence — that should be linked directly to the event.
superTaxon Taxon → Taxon	Identifies the parent taxon in the NCBI taxonomy hierarchy, enabling navigation up the taxonomic tree from species to kingdom.
surroundedBy † GO_CellularComponent → GO_CellularComponent	Represents the spatial containment (enclosed_by) relationship between GO Cellular Component terms, imported from the Gene Ontology — for example, the nucleoplasm is surroundedBy the nucleus.
tissue Cell, CellDevelopmentStep, or CellLineagePath → Anatomy	Associates a cell type or developmental event with the UBERON tissue term identifying the tissue context in which it occurs.
tissueLayer Cell → Anatomy	Associates a Cell instance with the UBERON tissue layer term identifying the specific tissue layer in which this cell type resides.
updatedInstance UpdateTracker → any Trackable class	Links a change record (UpdateTracker) to the specific curated instance — Event, PhysicalEntity, Regulation, etc. — that was changed at this release.

Reactome Data Model Glossary — Database-Derived Edition | V95

All descriptions grounded in live Neo4j database queries, informed by the V90 Reactome Data Model Glossary. 89 relationship types described in plain English.